7CS039-INYR-UM1: Summative Assessment 1

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1.0 Introduction

This report conducts an exploratory data analysis of a dataset focused on malignant melanoma survivors. The dataset comprises records of individuals who had surgery to treat malignant melanoma at Odense University Hospital in Denmark during the period from 1962 to 1977. The statistical computations and data visualizations were carried out using RStudio. Key variables identified in the dataset include ulceration, tumour thickness, and survival time, which were found to significantly influence mortality rates, with thicker tumours and ulcer presence being associated with higher mortality.

2.0 Data Overview: Dataset Summary and preprocessing

The dataset contains 205 entries with the following 8variables.Intialy on examining it was found that there were 8 variables, and one was redundant variable, and it was removed by dropping 1st column thus it became 7 variables. Then checked for null values and found dataset clean.

```
> #view structure of dataset
> str(melanoma_data)
'data.frame':   205 obs. of  7 variables:
 $ time     : int  10 30 35 99 185 204 210 232 232 279 ...
 $ status   : int  3 3 2 3 1 1 1 3 1 1 ...
 $ sex      : int  1 1 1 0 1 1 1 0 1 0 ...
 $ age      : int  76 56 41 71 52 28 77 60 49 68 ...
 $ year     : int  1972 1968 1977 1968 1965 1971 1972 1974 1968 1971 ...
 $ thickness: num  6.76 0.65 1.34 2.9 12.08 ...
 $ ulcer    : int  1 0 0 0 1 1 1 1 1 1 ...
> #view first few rows of dataset
> head(melanoma_data)
  time status sex age year thickness ulcer
1   10      3   1  76 1972      6.76     1
2   30      3   1  56 1968      0.65     0
3   35      2   1  41 1977      1.34     0
4   99      3   0  71 1968      2.90     0
5  185      1   1  52 1965     12.08     1
6  204      1   1  28 1971      4.84     1
```

```
> sum(is.na(melanoma_data))
[1] 0
> #drop 1st column
> melanoma_data <- melanoma_data %>% select(-1)
> head(melanoma_data)
  time status sex age year thickness ulcer
1   10      3   1  76 1972      6.76     1
2   30      3   1  56 1968      0.65     0
3   35      2   1  41 1977      1.34     0
4   99      3   0  71 1968      2.90     0
5  185      1   1  52 1965     12.08     1
6  204      1   1  28 1971      4.84     1
```

Code Snippet 1&output: Data overview & preprocessing

2.1 Summary Statistics

```
> #summary statistics
> summary_stats <- summary(melanoma_data)
> print(summary_stats)
      X              time           status            sex              age              year     
 Min.   :  1   Min.   :  10   Min.   :1.00   Min.   :0.0000   Min.   : 4.00   Min.   :1962  
 1st Qu.: 52   1st Qu.:1525   1st Qu.:1.00   1st Qu.:0.0000   1st Qu.:42.00   1st Qu.:1968  
 Median :103   Median :2005   Median :2.00   Median :0.0000   Median :54.00   Median :1970  
 Mean   :103   Mean   :2153   Mean   :1.79   Mean   :0.3854   Mean   :52.46   Mean   :1970  
 3rd Qu.:154   3rd Qu.:3042   3rd Qu.:2.00   3rd Qu.:1.0000   3rd Qu.:65.00   3rd Qu.:1972  
 Max.   :205   Max.   :5565   Max.   :3.00   Max.   :1.0000   Max.   :95.00   Max.   :1977  
   thickness         ulcer       
 Min.   : 0.10   Min.   :0.000  
 1st Qu.: 0.97   1st Qu.:0.000  
 Median : 1.94   Median :0.000  
 Mean   : 2.92   Mean   :0.439  
 3rd Qu.: 3.56   3rd Qu.:1.000  
 Max.   :17.42   Max.   :1.000  
```

	Variable	Min	1st Quartile (Q1)	Median	Mean	3rd Quartile (Q3)	Max
0	Time	10.000000	1525.000000	2005.000000	2153.000000	3042.000000	5565.000000
1	Status	1.000000	1.000000	2.000000	1.790000	2.000000	3.000000
2	Sex	0.000000	0.000000	0.000000	0.385000	1.000000	1.000000
3	Age	4.000000	42.000000	54.000000	52.460000	65.000000	95.000000
4	Year	1962.000000	1968.000000	1970.000000	1970.000000	1972.000000	1977.000000
5	Thickness	0.100000	0.970000	1.940000	2.920000	3.560000	17.420000
6	Ulcer	0.000000	0.000000	0.000000	0.439000	1.000000	1.000000

Code Snippet 2 & Table: indicating summary statistics

1.Time The survival time is measured in days it is a continuous variable which have a median of 2005 that is fifty percentage of patients lived over 5.5 years after surgery. The range is (10–5565 days) shows significant variation due to factors like tumour type and patient health. The mean is higher compared to median showing a distribution that is skewed to the right, with some individuals living far longer.

2. **Status (Patient Outcome)** Found to be a categorical variable indicating a binary outcome. The median status is 2 showing majority of patients were alive at the end of study. The mean value (1.79) show that a significant number of patients either lived or passed away from unrelated causes with fewer deaths directly attributable to melanoma.

3. Sex (0 = Female, 1 = Male) The median value of 0 indicate that most of the patients were female. The mean (0.385) confirms that males comprised approximately 38.5% of the sample.

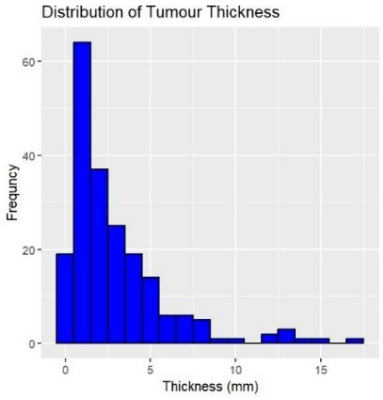
4. Age (in Years) The dataset reveals a median age of 54 years, meaning that this age was exceeded by half of the cases. The age range varies from 4 to 95 years showing a diverse age group, with most entries likely representing adults. The mean age of 52.46 years is close to the median, reflecting a nearly symmetrical variation of age within dataset.

5. **Year (Year of Operation)** Represents the year of observation or study. The operations span 15 years from 1962 to 1977.The median year is 1970 showing that 50 percentage of the surgeries occurred before this year.

6. **Thickness (Tumour Thickness in mm)** This variable is continuous and represents a measurable quantity. The median thickness of tumour is 1.94 mm, indicating that most tumours are relatively thin. The mean value of 2.92 mm exceeds the median, reflecting a right-skewed distribution caused by the presence of some very thick tumours. Tumour thickness serves as a significant prognostic indicator, as greater thickness may be associated with poorer outcomes. The noticeable skewness is evident from the higher mean compared to the median.

7. **Ulcer (Ulceration Indicator, 0 = Absent, 1 = Present)** A binary variable showing whether ulcer is there. The median value of 0 indicates that ulceration was absent in most cases. The mean (0.439) suggests that ulceration was present in approximately 43.9% of patients which could have a significant impact on survival. The dataset provides information on factors influencing survival after melanoma surgery. Key variables like tumour thickness and ulceration status show variability that could be correlated with survival outcomes. Analysis such as survival analysis or regression modelling could help explore these relationships in more depth. The mean of 0.439 suggests that ulcers are present in about 44% of cases, while the rest do not have ulcers.

2.2 Graphical Summary

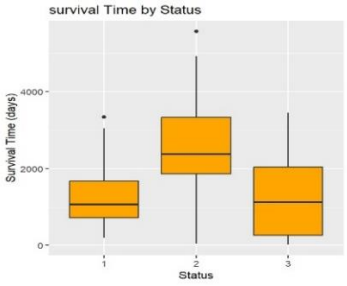



```
> #histogram of toumour thickness
> ggplot(melanoma_data,aes(x=thickness))+
+   geom_histogram(binwidth = 1,fill = "blue",color = "black")+
+   labs(title = "Distribution of Tumour Thickness",x = "Thickness (mm)",y = "Frequncy")
> #Boxplot survival time by status
> ggplot(melanoma_data,aes(x = factor(status),y = time))+
+   geom_boxplot(fill="orange")+
+   labs(title = "survival Time by Status", x = "Status", y = "Survival Time (days)")
> #Regression time and thickness
> model1 <- lm(time ~ thickness,data = melanoma_data)
> summary(model1)
```

Code Snippet3 & figure histogram for distribution of thickness

Observations: The distribution is **right skewed** meaning most tumours have smaller thickness values and a few tumours have significantly larger thicknesses. The majority of tumour thickness values are concentrated below 5 mm with a sharp decline in frequency as thickness increases. This skewness aligns with the summary statistics, where the mean thickness (2.92 mm) is higher than the median (1.94 mm), indicating a few extreme values are pulling the average upward. **Implications**: The skewness suggests that most patients have relatively thin

tumours, but a small subset of patients has much thicker tumours, which could indicate more severe cases.




```
> #histogram of toumour thickness
> ggplot(melanoma_data,aes(x=thickness))+
+   geom_histogram(binwidth = 1,fill = "blue",color = "black")+
+   labs(title = "Distribution of Tumour Thickness",x = "Thickness (mm)",y = "Frequncy")
> #Boxplot survival time by status
> ggplot(melanoma_data,aes(x = factor(status),y = time))+
+   geom_boxplot(fill="orange")+
+   labs(title = "survival Time by Status", x = "Status", y = "Survival Time (days)")
> #Regression time and thickness
> model1 <- lm(time ~ thickness,data = melanoma_data)
> summary(model1)
```

Code Snippet 4 & figure boxplot time by status

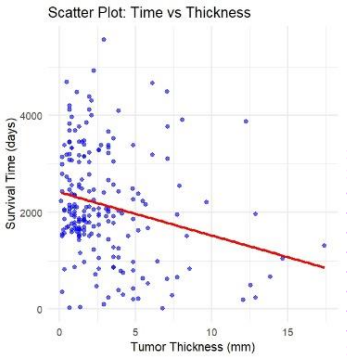
Observations: For **status = 1** (e.g., alive), survival times are generally higher, with a median around 3000 days. There are some outliers with lower survival times. For **status = 2** (e.g., deceased), survival times are lower, with a median around 1000 days and a narrower range. The variability in survival times is greater for patients who are alive compared to those who are deceased. Patients who are alive tend to have significantly longer survival times compared to those who are deceased. The outliers in the "alive" group could represent individuals with exceptionally low survival times despite their status.

Emerging Trends and Commentary:

1. **Tumour Thickness**: The right-skewed distribution suggests that while most tumours are thin, a few thick tumours may have clinical significance. Further investigation could explore how tumour thickness correlates with survival time or other variables.
2. **Survival Time by Status**: The clear distinction in survival times between the two status groups highlights the impact of survival status on duration. The presence of outliers, particularly in the "alive" group, may indicate cases that deviate from typical patterns, warranting further analysis.

The graphical summaries provide a deeper understanding of the dataset emphasizing the variability in tumour thickness and the relationship between survival time and status. This information could guide further exploration, such as identifying predictors of survival time or assessing the clinical significance of tumour thickness.

3.0 Regression and Correlation




```
> # Scatter plot with regression line: Time ~ Thickness
> plot1 <- ggplot(melanoma_data, aes(x = thickness, y = time)) +
+   geom_point(color = "blue", alpha = 0.6) +
+   geom_smooth(method = "lm", color = "red", se = FALSE) +
+   labs(
+     title = "Scatter Plot: Time vs Thickness",
+     x = "Tumor Thickness (mm)",
+     y = "Survival Time (days)"
+   ) +
+   theme_minimal()
> # Scatter plot with regression line: Time ~ Age
```

3.1 Regression (time-thickness)

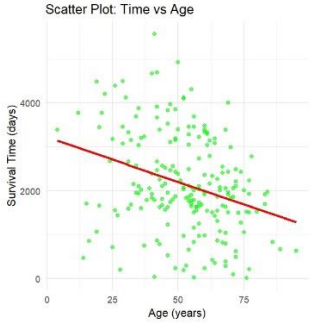
Parameter	Estimate	Remark
Intercept	2413.41	Predicted survival time when tumour thickness is 0.
Thickness Coefficient	-89.25	Survival time decreases by 89.25 days per 1 mm increase in thickness.
R-squared	0.05542	Tumour thickness explains only 5.54% of the variance in survival time.
Residual Standard Error	1093	High residual standard error suggests substantial unexplained variation.
F-statistic	11.91	Indicates a statistically significant relationship between time and thickness.
p-value	0.000679	The thickness coefficient is statistically significant at p < 0.001.

Code snippet 5 and Table 1: Regression for Time and Thickness

When tumor thickness is zero, the estimated survival time is about 2413 days. For every 1 mm increase in tumor thickness, survival time decreases by roughly 89 days, indicating that thicker tumors are linked to shorter survival times. This relationship is strongly supported by the p-value (< 0.0007).

A scatter plot shows a slight negative correlation between tumor thickness and survival time, with the regression line sloping downward. However, the low R^2 value (5.54%) suggests that tumor thickness explains only a small part of the variation in survival times. While thicker tumors are an important factor, other factors also significantly influence survival outcomes.

3.2 Regression (time-age)




```
> # Scatter plot with regression line: Time ~ Age
> plot2 <- ggplot(melanoma_data, aes(x = age, y = time)) +
+   geom_point(color = "green", alpha = 0.6) +
+   geom_smooth(method = "lm", color = "red", se = FALSE) +
+   labs(
+     title = "Scatter Plot: Time vs Age",
+     x = "Age (years)",
+     y = "Survival Time (days)"
+   ) +
+   theme_minimal()
```

Parameter	Estimate	Remark
Intercept	3217.45	Predicted survival time when age is 0.
Age Coefficient	-20.29	Survival time decreases by 20.29 days per 1 year increase in age.
R-squared	0.09091	Age explains only 9.09% of the variance in survival time.
Residual Standard Error	1072	High residual standard error suggests substantial unexplained variation.
F-statistic	20.3	Indicates a statistically significant relationship between time and age.
p-value	1.12E-05	The age coefficient is statistically significant at p < 0.001.

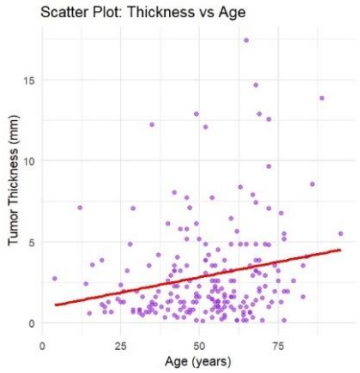
Code snippet 6 and Table 2: Regression for Time and Age

If age is zero (hypothetically), the predicted survival time is about 3217 days. For each additional year of age, survival time decreases by around 20 days on average. This shows that older patients generally have shorter survival times after surgery.

The scatter plot shows a weak negative relationship between age and survival time, with survival decreasing slightly as age increases. The regression analysis confirms a significant negative impact of age on survival, supported by a strong p-value (1.12e-05). However, the R^2 value (9.09%) suggests that age explains only a small part of the variation in survival times.

While age is an important factor in predicting survival outcomes, other factors also play a significant role. The model's reliability is supported by statistical tests, highlighting age as a key consideration in survival analysis.

3.3 Regression (thickness-age)




```
> # Scatter plot with regression line: Thickness ~ Age
> plot3 <- ggplot(melanoma_data, aes(x = age, y = thickness)) +
+   geom_point(color = "purple", alpha = 0.6) +
+   geom_smooth(method = "lm", color = "red", se = FALSE) +
+   labs(
+     title = "Scatter Plot: Thickness vs Age",
+     x = "Age (years)",
+     y = "Tumor Thickness (mm)"
+   ) +
+   theme_minimal()
> # Save plots
```

Parameter	Estimate	Remark
Intercept	0.94	Predicted tumour thickness when age is 0.
Age Coefficient	0.038	Tumour thickness increases by 0.038 mm per 1 year increase in age.
R-squared	0.04515	Age explains only 4.52% of the variance in tumour thickness.
Residual Standard Error	2.899	Moderate residual standard error suggests unexplained variation.
F-statistic	9.598	Indicates a statistically significant relationship between age and thickness.
p-value	0.00222	The age coefficient is statistically significant at $p < 0.01$.

Code snippet 7 and Table 3: Regression for Thickness and Age

When age is hypothetically zero, the estimated tumor thickness is about 0.94 mm. Each additional year of age increases tumor thickness by an average of 0.0377 mm. The p-value (0.00222) confirms a significant link between age and tumor thickness.

A scatter plot shows a slight positive trend, meaning tumor thickness tends to increase slightly with age. However, the R^2 value (4.51%) indicates that age explains only a small part of the variability in tumor thickness.

In summary, older individuals are slightly more likely to have thicker tumors, but other factors also play a major role.

3.4 Correlation

```
> # Correlation coefficients
> cor_time_thickness <- cor(melanoma_data$time, melanoma_data$thickness, use = "complete.obs")
> cor_time_age <- cor(melanoma_data$time, melanoma_data$age, use = "complete.obs")
> cor_thickness_age <- cor(melanoma_data$thickness, melanoma_data$age, use = "complete.obs")
> print(cor_time_thickness)
[1] -0.2354087
> print(cor_time_age)
[1] -0.3015179
> print(cor_thickness_age)
[1] 0.2124798
```

Variables	Correlation Coefficient	Interpretation
Time ~ Thickness	-0.235	Weak negative correlation: As tumor thickness increases, survival time tends to decrease slightly.
Time ~ Age	-0.302	Moderate negative correlation: Older patients tend to have slightly shorter survival times.
Thickness ~ Age	0.212	Weak positive correlation: Older patients tend to have slightly thicker tumors.

Code snippet 8 and Table 4: Correlation values for various parameters

Inferences Time and Thickness: Survival time decreases slightly as tumor thickness increases, shown by a weak negative correlation (-0.235), indicating a minor connection.

Time and Age: Older patients tend to have shorter survival times, with a moderate negative correlation (-0.302), suggesting age has a stronger effect than thickness.

Thickness and Age: Older patients are slightly more likely to have thicker tumors, indicated by a weak positive correlation (0.212), though the relationship is not strong.

4.Two Sample Significance Tests

A two-sample significance test checks if there is a real difference between two groups or if the difference is due to chance. A low p-value (below 0.05) suggests the groups are likely different, while a high p-value means they are similar. Here, we use two tests:

1. **Welch t-test**: Compares the average values (means) of the two groups.
2. **Wilcoxon test**: Examines the overall pattern (distribution) of values to identify differences between the groups.

```
> # T-test for 'thickness' by gender
> t_test_thickness <- t.test(thickness ~ sex, data =  melanoma_data)
> print(paste("Thickness - P-value:", t_test_thickness$p.value))
[1] "Thickness - P-value: 0.01009264720438"
> # Wilcox test for 'thickness' by gender
> w_test_thickness <- wilcox.test(thickness ~ sex, data =  melanoma_data)
> print(paste("Thickness - P-value:", w_test_thickness$p.value))
[1] "Thickness - P-value: 0.00421339798486194"
```

Code snippet 9 : Thickness-gender

Thickness: The p-value for thickness by gender is 0.01009, which is less than 0.05. The Wilcoxon test gave a p-value of 0. 004213.This means that there is a statistically significant difference in thickness between males and females. The t-statistic is negative, which suggests that males have a lower thickness than females. Both p-values here are below 0.05. Since both tests gave low p-values the study concludes that there is a significant difference in thickness between men and women. The "null hypothesis" (no difference) was rejected.

t-test for survival time by gender

```
> # T-test for 'time' by gender
> t_test_time <- t.test(time ~ sex, data = melanoma_data)
> print(paste("Time - P-value:", t_test_time$p.value))
[1] "Time - P-value: 0.038684472168065"
> # Wilcox test for 'time' by gender
> w_test_time <- wilcox.test(time ~ sex, data = melanoma_data)
> print(paste("Time - P-value:", w_test_time$p.value))
[1] "Time - P-value: 0.0404615104887267"
```

Code snippet 10: Time by gender

Two different statistical tests used to compare a variable called "time" between two groups (males and females). Both tests resulted in p-values smaller than 0.05 (0.03868 for the t-test and 0.04046 for the Wilcoxon test). A p-value smaller than 0.05 means the difference observed is statistically significant. Since the p-values are below 0.05, we reject the null hypothesis. The findings suggest that gender may influence the "time" variable in some way.

t-test age by gender

```
> # T-test for 'age' by gender
> t_test_age <- t.test(age ~ sex, data =melanoma_data)
> print(paste("Age - P-value:", t_test_age$p.value))
[1] "Age - P-value: 0.340771790333466"
> # Wilcox test for 'age' by gender
> w_test_age <- wilcox.test(age ~ sex, data = melanoma_data)
> print(paste("Age - P-value:", w_test_age$p.value))
[1] "Age - P-value: 0.34655399590921"
```

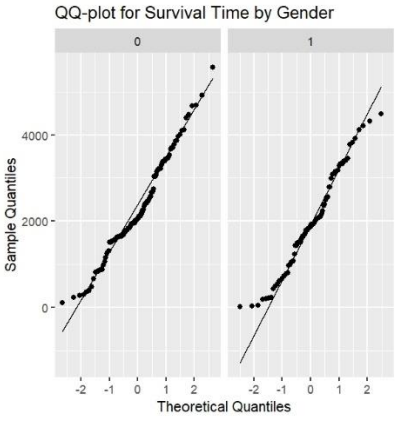
Code snippet 11: Age by gender

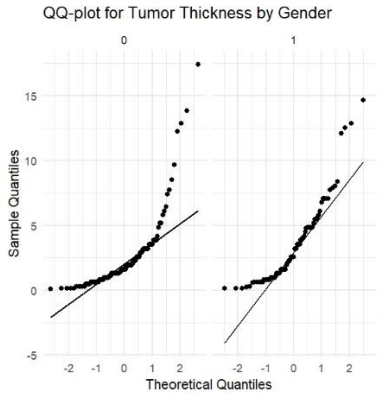
Age: The p-value for age by gender is 0.3408, which is greater than 0.05. This means that there is no statistically significant difference in age between males and females. The t-test gave a p-value of 0.3408 and Wilcoxon test gave a p-value of 0.3466. Since both p-values are above 0.05, the difference in age between men and women is not statistically significant. Age is not significantly impacted by gender. However, the study notes that tumour thickness and survival time are influenced by gender, based on other analyses.

Conclusion: Overall Results suggest that there are some statistically significant differences between males and females in the study population.

5 Q-Q Plots

QQ-Plots is used to check if the data for time, age, and thickness is spread out in a normal way. comparing men and women to see if gender influences these health-related factors and checking if the data follows expected patterns.








```
> # QQ-plot for time by gender
> ggplot(melanoma_data, aes(sample = time)) +
+   stat_qq() +
+   stat_qq_line() +
+   facet_wrap(~ sex) +
+   ggtitle("QQ-plot for Survival Time by Gender") +
+   xlab("Theoretical Quantiles") +
+   ylab("Sample Quantiles")
```

```
> # QQ-plot for thickness by gender
> ggplot(melanoma_data, aes(sample = thickness)) +
+   stat_qq() +
+   stat_qq_line() +
+   facet_wrap(~ sex) +
+   ggtitle("QQ-plot for Tumor Thickness by Gender") +
+   xlab("Theoretical Quantiles") +
+   ylab("Sample Quantiles") +
+   theme_minimal()
```

```
> # QQ-plot for age by gender
> ggplot(melanoma_data, aes(sample = age)) +
+   stat_qq() +
+   stat_qq_line() +
+   facet_wrap(~ sex) +
+   ggtitle("QQ-plot for Age by Gender") +
+   xlab("Theoretical Quantiles") +
+   ylab("Sample Quantiles") +
+   theme_minimal()
```

Code snippet 12: time by gender and qqplot **Code snippet 13:** thickness by gender and qqplot **Code snippet 14:** Age by gender and qqplot.

Time by Gender: Survival times are not normally distributed for both genders, with notable deviations and outliers, suggesting potential gender-based differences in survival patterns.

Tumour Thickness by Gender: Tumour thickness shows a skewed distribution for both genders with heavy tails, indicating the presence of extreme cases and a lack of normality in the data.

Age by Gender: Age is closer to a normal distribution for both genders, with minor deviations at the extremes, indicating that age is less influenced by non-normal factors compared to survival time and tumour thickness.

6.0 Critical aspects captured during the data exploration:

Tumour Thickness and Survival Time: There is a significant relationship between tumour thickness and survival time, emphasizing the need for early detection strategies to improve patient outcomes.

Gender-Specific Differences: Gender influences both tumour thickness and survival time, suggesting the need for gender-specific considerations in treatment planning.

Impact of Early Detection: Early detection strategies have the potential to significantly improve survival rates and overall patient outcomes in melanoma cases.

Evolution of Surgical Practices (1962–1977): The even distribution of surgeries over this period reflects consistent improvements in medical technology and standardization of procedures.

Temporal Insights on Melanoma: The data suggests a gradual advancement in surgical methods, contributing to better patient care over the analysed years.

Role of Monitoring and Research: Frequent monitoring and continuous research are essential for adapting to emerging patterns in melanoma care.

Adaptability in Healthcare: An adaptive approach to new trends ensures the development of effective preventive measures and personalized treatments.

Gender-Based Interventions: Addressing gender-related variations can enhance the impact of medical interventions and predictive models for melanoma patients.

Importance of Multidimensional Analysis: Exploring multiple factors, including gender, tumour characteristics, and time, provides a holistic understanding of melanoma outcomes.

Standardization of Procedures: Over the 15-year period, improvements in surgical techniques reflect the gradual standardization of melanoma treatments.

Healthcare Advancements: Continuous improvements in medical technologies during this period have contributed to consistent care delivery.

Insights into Melanoma Characteristics: The analysis sheds light on key characteristics that influence survival outcomes, aiding clinical decision-making.

Preventive Care Recommendations: Findings highlight the importance of preventive care, focusing on reducing tumour thickness through early interventions.

Personalized Treatment Plans: Personalized approaches based on individual and gender-specific data enhance treatment effectiveness and patient care.

Even Data Distribution: The even distribution of data over the years strengthens the reliability of the analysis and its applicability to clinical practice.

Enhanced Predictive Models: Incorporating gender-specific data and tumour characteristics improves the accuracy of survival predictions.

Improved Patient Care Strategies: The insights encourage the development of tailored patient care strategies that address specific risk factors.

Multifaceted Nature of Melanoma: Understanding the diverse factors affecting melanoma is key to improving treatment outcomes and survival rates.

Importance of Continuous Learning: The study underlines the need for healthcare practitioners to remain observant to changing trends and evolving practices.

Collaborative Efforts in Melanoma Research: A combined, interdisciplinary approach is crucial for advancing melanoma care and achieving better patient outcomes.

7.0 Future recommendations:

Enhance Data-Driven Personalization: Develop advanced predictive models using AI and machine learning to integrate tumour thickness, gender, and other factors, enabling personalized treatment and early intervention strategies.

Expand Screening Programs: Introduce targeted screening initiatives, especially in high-risk populations, to detect melanoma at its earliest stages and reduce tumour thickness-related complications**.**

Gender-Focused Research: Conduct further studies on gender-specific variations in melanoma progression to refine treatment protocols and explore hormonal or genetic factors that may influence outcomes.

Technology-Driven Surgical Innovations: Leverage modern technologies such as robotics and precision surgery to further standardize and improve melanoma surgical practices.

Global and Longitudinal Studies: Collaborate internationally to build longitudinal datasets, capturing trends over time to better understand melanoma's evolving nature and treatment efficacy across diverse populations.

8.0 References

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